

Custom Attributes

Custom attributes make it possible to extend an Entity, in order to add specific information supporting specific process needs in the customer solution. Once defined and published, Custom Attributes may be added to pages in IFS Cloud Web and IFS Cloud Mobile using the Page Designer.

Types of Custom Attributes

There are two main types of Custom Attributes.

- **Persistent** attributes are used to store extra information for an entity. These can be simply types like strings and numbers, enumerations, or a reference to another entity.
- **Read Only** attributes are used to look up related information (e.g. by following a reference) or to perform calculations on other attributes in the entity.

Performance Considerations

There is no specific upper limit on the number of fields that can be added to an entity.

Adding more attributes will not affect performance of the core business logic, but there will be a performance overhead when the custom attributes are added to / shown in pages, reports etc.

Generally speaking the overhead for persisted attributes is small, whereas for read only calculated attributes it can be larger depending on what exactly is calculated. So don't add more custom attributes than needed, and only add the ones really needed to be shown/edited on a page to that page.

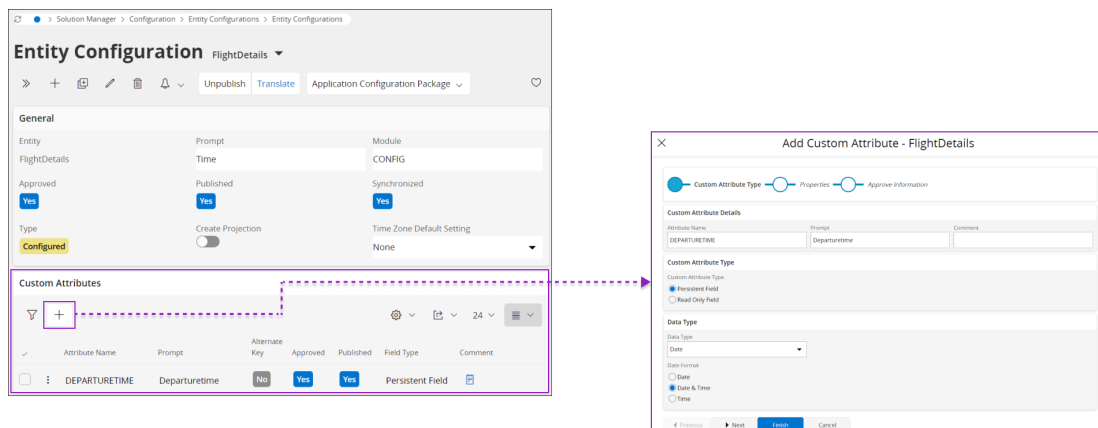
Custom Attributes Support in Structure-based Projection Actions

Structures can be defined using entities that come with default (nested) CRUD support. Unbound projection actions, which rely on such structures, also support custom attributes. To enable custom attribute support, these structures must be annotated with the **config entity** property. This functionality is particularly useful for enhancing integration APIs, including IFS premium APIs.

Create a Custom Attributes

Steps to follow for all Custom Attributes

1. Go to the **Entity Configuration** page in *Solution Manager > Configuration > Entity Configurations*.
2. Select **Configuration Entity** for which you want to create a Custom Attribute. You can either create a new Entity or search for an existing Entity.
3. In the Custom Attributes section, add a new record to start the 'Add Custom Attribute' assistant to configure the custom attribute to your liking. For more details on different types, refer to [Types of Custom Attributes](#).



4. By default only the main view for an Entity will be extended with the new custom attribute. In some cases the Entity may have multiple views used by different application pages. In these cases you have to approve the relevant detail views manually.
 - a. In the header section, follow the link **Approved Detail Views**
 - b. In the Approved Detail Views dialog, select in which additional views you want the custom attributes to appear.
5. [Approve](#) at least one of the attributes and the Entity.
6. [Publish](#) the Entity.
7. Optionally add Translations to your attributes.

Note: If you create a **Standard Entity with Timezone awareness** and subsequently create a Date/Time type custom attribute within it, the attribute will inherently possess timezone support.

Add Custom Attribute - FlightDetails

Custom Attribute Type — Properties — Approve Information

Custom Attribute Details

Attribute Name	Prompt	Comment
DEPARTURETIME	Departuretime	

Custom Attribute Type

Custom Attribute Type

☒ Persistent Field

☐ Read Only Field

Data Type

Data Type

Date

Date Format

☐ Date

☒ Date & Time

☐ Time

Previous Next Finish Cancel

Add Custom Attributes to Application Pages

After your custom attributes are published, they can be added to application pages.

1. Go to the page where you want to show the attribute.
2. Open the Page Designer and configure the page according to your preference.

Custom Attributes Support in Structure-based Projection Actions

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Find page

Configuration / Entity Configurations

Entity Configurations

Entity Configurations

Custom Attributes

Remaining Configuration Item Tables

ExtCustSched

ExtCustSchedPack

ExtCustSchedLine

ExtCustSchedPrevDeliv

Search

Advanced

Entity: ExtCustSched

Type

Module

More

Favorites

Search

Clear

Saved Searches

Entity Configuration

ExtCustSched

1 of 4

Unpublish

Translate

Additional Views

General

Entity: ExtCustSched

Prompt: Ext Cust Sched

Module: CUSCH

Approved: Yes

Published: Yes

Synchronized: Yes

Ready to Activate: Yes

Type: Standard

Custom Attributes

Attribute Name

Prompt

Approved

Published

Field Type

Comments

No of Transactions

Entity Reference

LDI view

Expression

Data Type

Format

Data Length

Insertable

DELIVERY_POINT_ADDR

Delivery Point Addr

Yes

Yes

Persistent Field

0

String

Unformatted

100

Yes

EXT_APPROVAL_DATE

Ext Approval Date

Yes

Yes

Persistent Field

0

Date

Date & Time

Yes

EXT_APPROVAL_QTY

Ext Approval Qty

Yes

Yes

Persistent Field

0

Number

Unformatted

Yes

EXT_APPROVAL_RECEI...

Ext Approval Received

Yes

Yes

Persistent Field

0

ExtBoolean

Enumeration

Unformatted

50

Yes

Find page

Configuration / Entity Configurations

Entity Configurations

Entity Configurations

Custom Attributes

Remaining Configuration Item Tables

ExtCustSched

ExtCustSchedPack

ExtCustSchedLine

ExtCustSchedPrevDeliv

Search

Advanced

Entity: ExtCustSched

Type

Module

More

Favorites

Search

Clear

Saved Searches

Entity Configuration

ExtCustSchedLine

3 of 4

Unpublish

Translate

Additional Views

General

Entity: ExtCustSchedLine

Prompt: Ext Cust Sched Line

Module: CUSCH

Approved: Yes

Published: Yes

Synchronized: Yes

Ready to Activate: Yes

Type: Standard

Custom Attributes

Attribute Name

Prompt

Approved

Published

Field Type

Comments

No of Transactions

Entity Reference

LDI view

Expression

Data Type

Format

Data Length

Insertable

LOCATION_ADDR

Location Addr

Yes

Yes

Persistent Field

0

String

Unformatted

100

Yes

QTY_APPROVAL_DATE

Qty Approval Date

Yes

Yes

Persistent Field

0

Date

Date & Time

Yes

QTY_APPROVED

Qty Approved

Yes

Yes

Persistent Field

0

Number

Unformatted

Yes

Search...

Service Operations - Actions

Invoke action
ReceiveCustomerSchedule

Service Operations - Functions

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Invoke action ReceiveCustomerSchedule

This operation is used to receive delivery plans sent from the customer to the supplier.

AUTHORIZATIONS: > (OpenId and basicAuth)

REQUEST BODY SCHEMA: application/json

request body for action ReceiveCustomerScheduleActionImport

ExtCustSchedStruct

object

A structure of type ExtCustSchedStruct which contains information of External Customer Schedule

CrossRefNo	string <= 35 characters This field is a unique identifier used to enter multiple Customer Schedules with the same data set. This field is mandatory if the connected customer schedule agreement part is linked to the customer schedule agreement that contains any customer agreement cross references. Entered value of the customer agreement cross reference in the customer schedule is validated against the value in the relevant customer schedule agreement.
DocNo required	string <= 2000 characters The unique identity given for the message.
ValidFrom required	string <date> This value indicate from what date the schedule is valid.
MessageType	string <= 2000 characters The message type of the customer schedule. This is a free text field. The intended use is to provide the possibility for a customer to include the specific naming of a message according to the EDI standard the customer is using. Optionally, an EDI converter can provide a value for this field. Example: If using Edifact, the message type can be set to DELFOR(Plan) and DELJIT(Calloff). If using X12 the type could be 830 (Plan) and 862 (Calloff)
SupShipFromInfo	string <= 2000 characters Supplementary information from the customer about the location from which the parts are to be delivered. This can be, for example, the identification number of a factory or factory area.
SupShipToInfo	string <= 2000 characters Supplementary information from the customer about the destination to which the parts are to be delivered. This can be, for example, the identification of a factory or factory area.
ScheduleType required	string Enum: "Plan" "Calloff" This field indicate whether the schedule is a plan or a call-off.
CustOwnDelAddr required	string <= 2000 characters This is the deliver-to-customer's own identification of the delivery address. This ID is used to recognize addresses between sites and companies in EDI communication as well as in inter-site order communication. This ID is sometimes referred to as the EAN location number.
Customer	string <= 2000 characters The short name or identification number of the ordering customer.
ValidTo required	string <date> This value indicate to what date the schedule is valid.
ArticleLines >	Array of objects (ExtCustSchedStructArticleLine) Include header and part information of an incoming customer schedule record in IFS Cloud.
ExtCustSchedLines >	Array of objects (ExtCustSchedStructExtCustSchedLine) Include schedule lines that connects to an incoming customer schedule record in IFS Cloud.

Array (

NoteText	string <= 2000 characters This is a free-text field up to 2000 characters, that is linked to the schedule lines of the incoming message.
RefId	string <= 2000 characters The customer-provided reference ID for a specific demand. Reference ID can only be used if the agreement part for the schedule allows the use of references.
DockCode	string <= 2000 characters It is used to specify delivery information and define an area (dock) where the parts are to be delivered, i.e., a cargo bay.
SubDockCode	string <= 2000 characters It is used to specify delivery information and define an area on the dock where the parts are to be delivered, i.e., a gate in the cargo bay.
PatternDescription	string <= 2000 characters Information from the customer about the time frame a specific demand quantity covers. This can be, for example, Daily, Weekly, Monthly, or Bi Monthly.
PortOfLoading	string <= 2000 characters Information from the customer about the location at which the parts are to be loaded. This can be, for example, the identification number of a shipping dock inside a factory or a factory area.
ManufacturingDepartment	string <= 2000 characters The manufacturing department to which the delivery should be made, as specified by the customer. This information is printed on a delivery note that is used when the delivery is received at the customer site, i.e., the receiver then knows to which department this specific delivery is intended.
PartPrice	number Information received from the customer regarding the price of a part. The price is usually expressed in the customers default currency. The value of the Use Contract option in the same message decides whether the price is used. If the indicator is not selected (or the information is not received) the received price is used on the customer order lines created upon the release of the customer schedule. If the option is selected, the normal price retrieval logic decides the price independently of the part price received in the message. Typically, a price is included in a customer schedule when price cuts occur. The customer and the supplier have agreed upon the size of the price cut in advance, but not the date from which the new price will be valid.
DeliverySequence	number Information received from the customer regarding the order in which the deliveries should be made. Typically, this information is received for firm demands with the same Wanted Delivery Time.
ToDate	string <date> Each demand in a customer schedule is intended to cover the customers needs within a date range. The boundaries of the range is set by the From Date and To Date stated for each demand. The To Date can be a non working day but must be a later date than the date entered as the From Date.
UseContract	boolean Information from the customer regarding the price that should be used when releasing a customer schedule and creating

	customer with the schedule regarding the price that should be used when creating a customer schedule and creating customer order lines. The values can be True or False. True indicating that the price received in the customer schedule should not be used. In this case the normal price retrieval logic decides the price. False indicating that the price received in the schedule should be used. If no value is received its interpreted as False.
ProvisionalPrice	boolean Information received from the customer. The provisional price indicator tells if the part price received in the same schedule is provisional or fixed. A provisional price can later be updated by the customer. If the part price in the message is used, the value of the provisional price indicator will be passed on to the customer order line created upon releasing of the schedule.
ScheduleNo	number This field is used for indicating the number or identification of the schedule.
CustPartNo	string <= 2000 characters The customer's part number corresponding to the sales part number in IFS Cloud.
LineType	string <= 2000 characters This column indicate whether the schedule line should be delivery authorized, i.e., whether or not the line should be converted into a customer order line. To convert the line into a customer order line, you must also have indicated on the agreement that the schedule type is allowed to create purchase order lines.
ToLocationNo	string <= 2000 characters The location number of the destination location for the part at the customer site.
CustomerPoNo	string <= 2000 characters The customer's purchase order number. This number can be sent in a customer schedule with the intention that it should work as a reference for a specific demand, together with the purchase order line number, purchase order release number, and the reference ID. The intention is also to create a one to one relation between the customers purchase order and the suppliers corresponding customer order. The customer will, in later discussions, refer to his own purchase order number when having questions regarding the demands communicated on the customer schedule. If the purchase order information is received in a customer schedule, IFS will pass the information to the customer order created at release of the schedule, to support the features described above. The customer purchase order number also appears as an order reference on any document sent to the customer. To be able to receive the purchase order information, the customer schedule agreement for the part must be set up to use references and to create customer orders on demand. The purchase order information is typically sent for delivery authorized demands, i.e., firm demands.
CustPoLineNo	string <= 2000 characters The purchase order line number as specified by the customer in the customer schedule. This works as a reference for a specific demand along with the purchase order line number, purchase order release number and a reference ID. In order to be able to receive the purchase order information the customer schedule agreement for the part must have been set up to use references and to create customer orders on demand. The purchase order information is typically sent for delivery authorized demands, i.e. firm demands.
CustomerPoRelNo	string <= 2000 characters The customer's purchase order number in the customer schedule; it can be used as a reference for a specific demand together with the purchase order line number, purchase order release number, and reference ID. If the purchase order information is received in a customer schedule, IFS Applications will pass the information to the customer order created at release of the schedule. To be able to receive the purchase order information, the customer schedule agreement for the part must be set up to use references and to create customer orders on demand. The purchase order information is typically sent for delivery authorized demands, i.e. firm demands.
CumQtyDue	number The quantity of parts that the customer wish to receive in total. If you enter the quantity manually, you can also choose to enter the qua data in the Qty Due column instead; then, the system will calculate and display the cumulative quantity. Note that the accumulation of quantity is only done within this single schedule.
DueQty required	number Demand quantity for this schedule line.
DeliveryDateFromDate required	string <date-time> The date when the customer wants this particular delivery. Depending on how lead times are defined in IFS, this can be either the delivery date or the customer receipt date.
Cf_Location_Addr	string <= 100 characters
Cf_Qty_Approval_Date	string <date-time>
Cf_Qty_Approved	number
}	
ExtCustSchedPrevDeliveries >	Array of objects (ExtCustSchedStructExtCustSchedPrevDeliv) Include information related to the previous deliveries made to the customer on an incoming customer schedule record in IFS Cloud.
ExtCustSchedPacks >	Array of objects (ExtCustSchedStructExtCustSchedPack) Include packing information connected to an incoming customer schedule record in IFS Cloud.
Cf_Delivery_Point_Addr	string <= 100 characters
Cf_Ext_Approval_Date	string <date-time>
Cf_Ext_Approval_Qty	number
Cf_Ext_Approval_Received	string (CfEnum_FndBoolean_Enumeration) Enum: "CfEnum_FALSE" "CfEnum_TRUE"

Custom Attributes in Outbound XML & JSON Messages

For structures implemented with default (nested) CRUD support, the custom attributes of the entities on which structure is based on, can be included in both **XML & JSON** type outbound messages.

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Note*: In order to include such custom attribute in a relevant outbound message, Enable on Outbound Messages* *property on the custom attribute configuration should be enabled.*

